

Modelling of Large-Scale Circulation in Atmosphere and Ocean
(MO8004)

Rossby Waves

Lab 3

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Questions

Theory

1. State the necessary assumptions that are needed to study Rossby waves.
2. Write down the dispersion relation of Rossby waves, the equation for the group velocity, and the phase speed. Discuss the dispersion of Rossby waves.
3. Discuss in your own words the term dispersion.
4. From the equations in Question 2, discuss the role of the wave number k in relation to the Rossby radius R . Which cases can be distinguished?
5. State the equation for $\partial h/\partial t$ with variable depth

$$D = D_0 + \alpha y$$

and indicate the two additional terms that need to be implemented.

6. State the relation where α cancels the β -effect.
7. Why do we have to assume reduced gravity in this laboratory?

Experiments

Set up the model as described in the laboratory description for Laboratory 3B and follow the instructions for the experiments.

2.1 The role of wave number

1. List six values of kR and the corresponding L_w that sample the characteristic shape of the zonal Rossby wave phase speed c_x and group speed c_{gx} relations, assuming

$$L_w = \frac{1.3\sqrt{g'H}}{kR\sqrt{5}f_0}.$$

Explain your choice.

2. Plot the theoretical zonal phase speed c_x and group speed c_{gx} as a function of kR , and highlight the values corresponding to your choices in Question 1.
3. Choose three kR values representing the characteristic Rossby wave regimes and provide Hovmöller diagrams at the center of the y -axis. Estimate the numerical zonal phase and group velocities and plot them as lines in the diagrams. Add the theoretical zonal phase and group speeds as lines in the Hovmöller diagrams.
4. Add the numerical estimates of your three characteristic waves into the plot from Question 2. (It is acceptable to combine Questions 2 and 4 into one final figure.)

2.2 The role of topography

1. Fix β and choose a value of kR for experiments with varying α . Plot Hovmöller diagrams of at least three representative waves on the α -plane. Add the theoretical zonal phase and group speeds to the plots and explain your parameter choices.
2. Plot the waves shown in Question 1 as xy cross-sections at an appropriate time.
3. For two of your α -value experiments (excluding the cancelling case), plot the theoretical c_g and c_x as a function of kR and include the numerical estimates from the experiments. Compare the results with the plot from Question 2.1.4.

1 Theory

v) The normal convergence-equation in the shallow water equations is

$$\frac{\partial \eta}{\partial t} + \frac{\partial(uh)}{\partial x} + \frac{\partial(vh)}{\partial y} = 0. \quad (1)$$

Normally we use that $h = H_0 + \eta$, where $\eta \ll H_0$. But now we instead use a varying topography with $h = H_0 + \alpha y + \eta$, where $\eta \ll H_0$ and $\alpha y \ll H_0$. Using this on Equation 1 we obtain

$$\Rightarrow \frac{\partial \eta}{\partial t} + H_0 \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) + \underbrace{\alpha y \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right)}_{\text{new topographic terms}} + \alpha v = 0. \quad (2)$$

2 Experiments

2.1 The rol of wave number

- i) only theoretical
- ii) Only theoretical
- iii) [Döös et al., 2022]

2.2 The role of topography

- i) Choose a kR where we see stuff, i.e. do not choose extremes.
- ii) Do a normal contour-plot. for the different α .
- iii)

A The Source Code

The source code for this lab can be found on GitHub:

`https://github.com/FredrikBergelv/
Modelling-of-Large-Scale-Circulation-in-Atmosphere-and-Ocean`

References

Kristofer Döös, Peter Lundberg, and Aitor Aldama Campino. *Basic Numerical Methods in Meteorology and Oceanography*. Stockholm University Press, Stockholm, Sweden, 2022. ISBN 978-91-7635-172-7. doi: 10.16993/bbs.